

Namra Desai

Curriculum Vitae - M.S. Artificial Intelligence Applicant

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OBJECTIVE

Software developer with a Master of Science in Computer Science and 4+ years of experience building distributed, high-throughput production systems, seeking admission to a Master's program with an Artificial Intelligence specialization. Motivated to build on a foundation in systems and backend engineering, along with self-directed study in computer vision, graph neural networks, and MLOps, to pursue rigorous graduate-level research in AI.

EDUCATION

University of North Carolina at Charlotte, USA *Charlotte, USA | Aug 2018 – May 2020*

Master of Science in Computer Science

- Served as Graduate Teaching Assistant for Algorithms and Data Structures under Dr. Qiong Cheng.
- Coursework included Algorithms & Data Structures, Visual Analytics, Intelligent system , Network Based Application Development with independent study extending into machine learning and applied AI following graduation.

AI/ML PROJECTS

Edge-Optimized Real-Time Multi-Object Tracking System | *Independent Project*

Python, YOLOv10, ByteTrack, TensorRT, FastAPI, Docker

- Built a computer vision pipeline to detect and track objects across live video streams with persistent ID retention, using YOLOv10 and ByteTrack.
- Explored model optimization for edge deployment by quantizing YOLOv10 from FP32 to INT8 precision with TensorRT, achieving a 3.4x inference speedup (15 → 52 FPS) with minimal accuracy loss.
- Containerized the inference pipeline with Docker and exposed it through a FastAPI backend supporting sub-50ms response times.

Financial Fraud Detection via Relational Graph Neural Networks | *Independent Project*

Python, PyTorch Geometric, NetworkX, GNNExplainer, Scikit-Learn

- Modeled a financial transaction dataset as a heterogeneous graph (100K+ nodes/edges) to capture structural fraud patterns missed by traditional ML approaches.
- Implemented a Relational Graph Convolutional Network (R-GCN) using PyTorch Geometric, improving minority-class F1-score by 14% on imbalanced fraud data.
- Applied GNNExplainer for post-hoc explainability, visualizing key structural paths contributing to fraud predictions for audit interpretability.

End-to-End MLOps & Automated Retraining Pipeline | *Independent Project*

Python, Apache Airflow, DVC, MLflow, EvidentlyAI, GitHub Actions

- Built an automated MLOps pipeline for continuous data ingestion, feature engineering, and model deployment for a demand-forecasting use case.
- Used Data Version Control (DVC) for dataset versioning and MLflow to track experiments and hyperparameter tuning across 50+ training runs.
- Set up real-time data drift monitoring with EvidentlyAI and automated GitHub Actions CI/CD workflows to trigger retraining and redeployment on performance degradation.

Asynchronous RAG Framework with Hybrid Retrieval & Semantic Caching | *Independent Project*

Python, FastAPI, Redis, JWT, Vector Embeddings

- Developed an asynchronous FastAPI RAG framework with a document ingestion pipeline using high-dimensional vector embeddings.
- Mitigated computational redundancy and API costs by designing a semantic caching layer utilizing Redis distance-metrics.
- Engineered a hybrid retrieval system combining sparse and dense embedding models to achieve 92% retrieval precision.
- Implemented formal access-control mechanisms incorporating token-bucket rate-limiting algorithms and cryptographic JWT authentication.

HACKATHONS & COMPETITIONS

- 2nd Place - Queen City Data Science Hackathon 2019, Red Ventures.
- 5th Place - Reinvent the Wheel Data Science Hackathon 2019, ATD (applied ML for a tire-industry business use case).
- 5th Place - Queen City Data Science Hackathon 2020, Red Ventures (data science solution addressing the opioid crisis).

National Smart India Hackathon 2017 - Selected Participant | 2017

- Designed and built a one-point secure digital document locker enabling students to store college admission and personal documents, and share access keys with universities for verification.
- Eliminated the need for physical document submission by enabling secure, permissioned digital access for institutional verification.

WORK EXPERIENCE

Software Developer, Crispin Corporation (Sony Group Company)

Raleigh, USA | Feb 2021 – Sept 2025

- Designed and implemented an end-to-end distributed monitoring platform (real-time data collection, multi-agent aggregation, operator-facing visualization) supporting real-time operations across 100+ sites.
- Optimized high-throughput event processing pipelines (100+ sites, ~5,000+ events/channel) through concurrency, caching, and batch processing, improving scalability and reliability.
- Developed distributed data synchronization and real-time broadcasting services, and led architectural improvements to reduce latency and improve user experience.

Freelance Software Developer, Valarix AI

Remote (Mexico-based client) | Feb 2026 – Present

- Built a production Python/Playwright automation platform for SAT (Mexican tax authority) document retrieval and e.firma/FIEL credential validation, integrating REST APIs, encrypted credential handling, and automated FTP delivery.
- Engineered parallel multi-customer processing with Python multiprocessing/ProcessPoolExecutor, running concurrent browser sessions while preserving per-customer sequential integrity and adding interrupted-run recovery for unattended reliability.
- Deployed and operated the system in production on a DigitalOcean Linux droplet via PM2, sustaining ~360–380+ document requests/hour while diagnosing and resolving CPU, memory, I/O, and timeout bottlenecks.

Software Developer Intern, Includiv

Cambridge, USA | Nov 2020 – Feb 2021

- Implemented new features with responsive layouts and optimized performance, ensuring seamless integration with existing code. (React.js, Hooks, TypeScript)

TECHNICAL SKILLS

Languages: Python, JavaScript, C#

AI/ML: PyTorch, PyTorch Geometric, Scikit-Learn, YOLOv10, TensorRT, MLflow, DVC, EvidentlyAI, Apache Airflow

Frameworks: Node.js, Express.js, Django, Flask, FastAPI

Tools & Infra: Docker, Git, GitHub Actions, RabbitMQ, SQLite, MySQL, PM2, DigitalOcean

Practices: Object-Oriented Design, RESTful APIs, Agile Development, SDLC